Public Electric Vehicle Charging Access in New Jersey

Spatial Statistics Project

May 5, 2026

1 Executive Overview

This project examines the spatial distribution of public electric vehicle (EV) charging stations across New Jersey census tracts. The main research question is:

> Are public EV charging stations in New Jersey evenly distributed across census tracts, or are they concentrated in certain socioeconomic and geographic areas?

The analysis combines three datasets: public EV charging station data, American Community Survey (ACS) demographic data, and U.S. Census tract boundary data. The unit of analysis is the census tract. Public charging stations are point locations, while ACS variables are tract-level attributes. Therefore, a spatial join is used to assign each charging station to the tract that contains it.

The project uses exploratory data analysis, maps, Moran's I, and negative binomial regression. The main findings are:

- New Jersey has 2,181 census tracts in the analysis boundary file.
- The filtered station dataset contains 1,822 public open EV charging stations.
- These stations represent 3,965 Level 2 ports and 2,062 DC fast ports, for a total of 6,027 ports.
- Only 701 of 2,181 tracts, or about 32.1%, contain at least one public charging station.
- EV charging access is zero-heavy and right-skewed: most tracts have no stations, while a smaller number contain several.
- Moran's I indicates modest but statistically significant spatial clustering in station access.
- Negative binomial regression suggests that median household income is positively associated with station counts, while owner-occupancy is negatively associated.
- A density-adjusted model improves model fit, but residual spatial autocorrelation remains, suggesting that important geographic predictors are still missing.

Notice that this analysis should be interpreted as exploratory rather than causal. Station placement seems to depends on factors not included in the current model, such as highway access, commercial land use, parking availability, tourism, employment centers, utility capacity, and EV adoption.

2 Study Area and Spatial Unit

The study area is New Jersey. The spatial unit is the census tract. A census tract is a small statistical geography designed by the U.S. Census Bureau to approximate neighborhood-level areas. **Census tracts are useful** because they are smaller than counties and allow a more detailed spatial analysis of neighborhood-level infrastructure access. Each row in the final analysis dataset represents one census tract. For each tract, the following was stored: demographic variables, charging station counts, charging port counts, per-capita charging rates, distance to the nearest charger, and model residuals.

3 Data Sources and Dataset Features

3.1 Public EV Charging Station Data

The EV charging station dataset provides point locations of public charging infrastructure. Each row represents a station location. A station here is a physical site, such as a parking lot, shopping center, highway rest stop, or office location. A station may contain multiple charging ports. A port is an individual connector that can charge a vehicle. Important variables include:

Variable	Meaning
`Station.Name`	Name or label of the charging location
`Street.Address`, `City`, `ZIP`	Address information
`Latitude`, `Longitude`	Coordinates used for mapping and spatial joins
`State`	Used to filter to New Jersey
`Access.Code`	Used to keep public stations
`Status.Code`	Used to keep open stations
`EV.Level2.EVSE.Num`	Number of Level 2 ports
`EV.DC.Fast.Count`	Number of DC fast ports
`EV.Network`	Charging network, such as ChargePoint, Tesla, or EVgo
`Facility.Type`	Type of place where station is located, if available
`Open.Date`	Date the station opened, if available

Table 1: Main EV station dataset features.

Mainly the use station locations, Level 2 port counts, and DC (Direct Current fast chargers)fast port counts. Other variables, such as network, connector type, facility type, access hours, and opening date, are not central to the current analysis but could be used in future work.

3.2 Census Tract Boundary Data

The tract boundary shapefile is for the polygon geometry for New Jersey census tracts. Where we are going to use for mapping, spatial joins, Moran's I, and tract-level aggregation. Important fields include:

Variable	Meaning
`STATEFP`	State FIPS code; New Jersey is 34
`COUNTYFP`	County FIPS code
`TRACTCE`	Census tract code within county
`GEOID`	Unique tract identifier used for joins
`ALAND`	Land area, useful for density calculations
`AWATER`	Water area
`geometry`	Tract polygon geometry

Table 2: Main census tract shapefile features.

The **GEOID** is especially important. It combines state, county, and tract identifiers, and **is the key used to join ACS demographic variables to tract polygons.**

3.3 ACS Demographic Data

The ACS dataset provides tract-level demographic and socioeconomic variables. These variables can help us to evaluate whether charging access is associated with neighborhood characteristics. Important ACS-derived variables include:

Variable	Meaning
`total_population`	Total tract population
`median_income`	Median household income
`poverty_rate`	Percent of population below poverty
`pct_no_vehicle`	Percent of households without vehicle access
`pct_owner_occupied`	Percent of occupied housing units that are owner occupied

Table 3: Main ACS variables used in the analysis.

Notice that current ACS variables are enough for the scope of the analysis. However, future work could include race and ethnicity, education, renter share, commute mode, multi-unit housing, and employment density.

4 Data Cleaning

4.1 ACS Cleaning

ACS API values are read as text, so they must be converted to numeric variables. The code also handles two common data issues:

- Census missing-value sentinels, especially negative median income values, are converted to `NA`.
- Percent variables are calculated only when their denominators are positive.

The key formulas are:

$$\text{poverty rate}_i = 100 \times \frac{\text{people below poverty}_i}{\text{poverty}_i}, \tag{1}$$

$$\text{no-vehicle share}_i = 100 \times \frac{\text{households without vehicle}_i}{\text{total households}_i}, \tag{2}$$

$$\text{owner-occupied share}_i = 100 \times \frac{\text{owner-occupied housing}_i}{\text{occupied housing}_i}. \tag{3}$$

During the cleaning the following was found: 31 negative median income values, 13 zero poverty denominators, 15 zero household denominators, and 15 zero housing denominators. These were handled by setting affected variables to missing.

4.2 EV Station Cleaning

The EV station dataset was filtered to:

- New Jersey stations only
- public access stations only
- open stations only
- stations with valid latitude and longitude

Missing Level 2 or DC fast port values were treated as zero for the relevant port type. This is decision was made because a missing Level 2 count at a DC fast station, for example, does not necessarily mean that the station is missing.

5 Spatial Join and Tract-Level Dataset Construction

Charging stations are point locations, while ACS data and tract boundaries are polygon data. To connect the two, the station points and tract polygons were projected to **EPSG 5070** before performing the spatial join. This because projected coordinate systems are more appropriate for spatial operations.

Each station point was assigned to the census tract polygon that contains it. The join assigned all 1,822 stations to tracts, so there was no un-match station locations.

After the spatial join, station counts and port counts were aggregated by tract:

$$\text{stations total} = \sum_{j=1}^{n_i} 1, \tag{4}$$

$$\text{ports total} = \text{Level 2 ports} + \text{DC fast ports}. \tag{5}$$

Tracts without any matched station were assigned zero stations and zero ports. This is not missing data, this was just interpreted as no public open station was present in the station file for that tract.

6 Rate Construction and Outlier Handling

The main population-adjusted access variable is stations per 10,000 residents:

$$\text{stations per 10k} = \frac{\text{stations total}}{\text{population}} \times 10,000. \tag{6}$$

Three versions of the rate are used:

Variable	Meaning
`stations_per_10k_raw`	Direct population-adjusted rate
`stations_per_10k_clean`	Rate after excluding tracts with population below 500
`stations_per_10k_map`	Cleaned rate capped at 75 for map display only

Table 4: Station-rate variables.

Very small-population tracts can make the station rate look misleading (high). For example, if a tract has only a few residents but one charging station, its stations-per-10,000-residents rate can become extremely large. That happened in the raw data: the highest raw rate was 27,500 stations per 10,000 residents. After removing tracts with fewer than 500 residents, the highest rate dropped to 97.29. This change affected only 16 tracts. Also, only one tract had to be capped at 75 on the map. The cap was used only to make the map easier to read. **This did not change the data used in the summaries or regression models.**

7 Exploratory Data Analysis

7.1 Statewide Summary

The statewide summary shows:

Metric	Value
Total New Jersey census tracts	2,181
Public open EV stations	1,822
Level 2 ports	3,965
DC fast ports	2,062
Total ports	6,027
Tracts with at least one station	701
Tracts without a station	1,480
Percent of tracts with at least one station	32.14%

Table 5: Statewide EV charging infrastructure summary.

The key interpretation of this is that public charging infrastructure exists in only about one-third of New Jersey census tracts. **This suggests that access is concentrated rather than evenly distributed.**

7.2 County-Level Summary

County-level EDA provides a familiar geographic scale before moving into detailed tract-level maps. The counties with the most public stations are Bergen, Middlesex, Essex, Hudson, and Morris. Bergen has 239 stations, Middlesex has 209, Essex has 164, Hudson has 141, and Morris has 132.

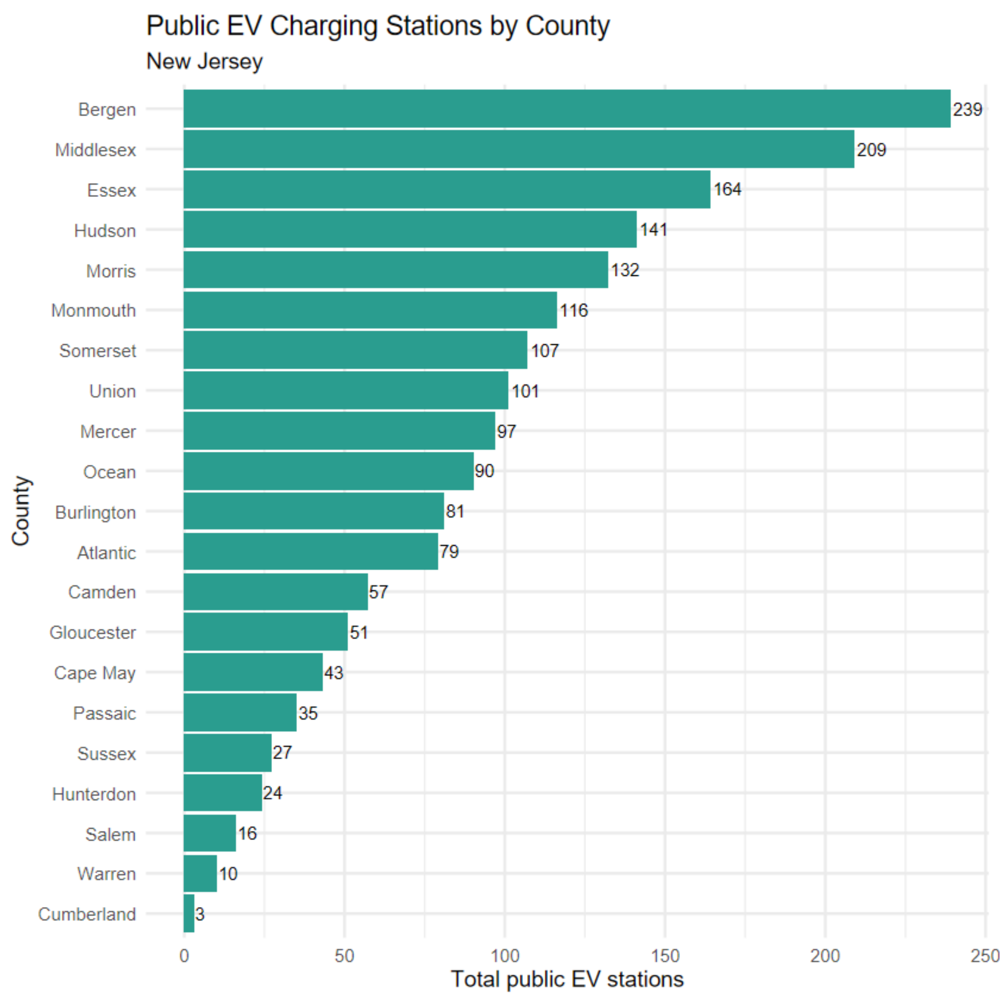

Figure 1: Public EV charging stations by county. Longer bars represent higher station counts.

7.3 Distribution of Station Counts by Tract

The distribution of station counts is zero-heavy and right-skewed. Most tracts have no station, while a smaller number contain one or more stations. To make this easier to read, the distribution can be displayed with two separate plots: one for tracts with 0–10 stations and one for tracts with more than 10 stations.

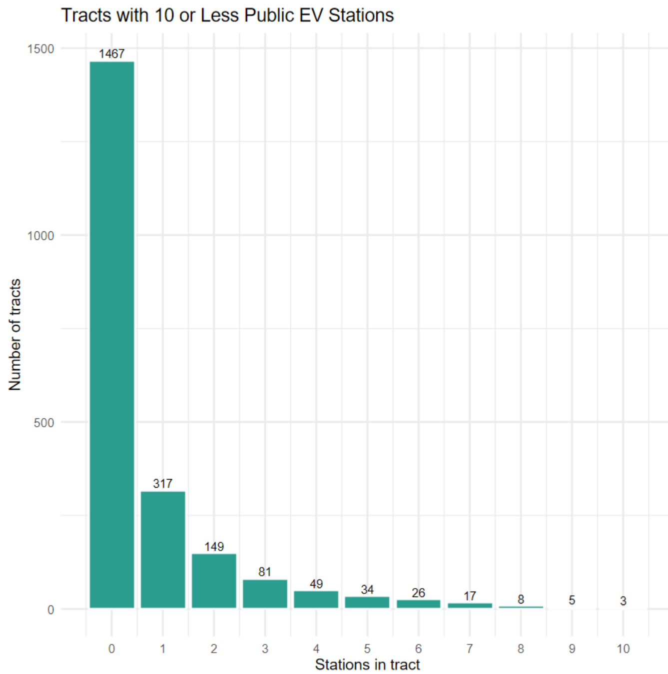

Figure 2: Distribution of public EV station counts for tracts with 0–10 stations. The large zero-station bar shows that most tracts have no public EV station.

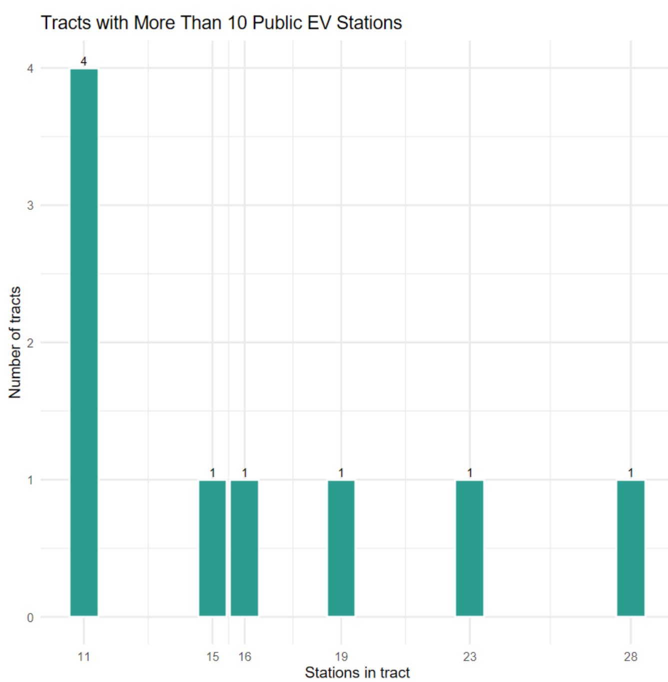

Figure 3: Distribution of public EV station counts for tracts with more than 10 stations. These high-count tracts are rare and likely represent commercial, workplace, highway, or destination charging clusters.

This distribution supports the use of a count model rather than ordinary linear regression. The outcome is nonnegative, integer-valued, zero-heavy, and right-skewed.

7.4 Station Rates per 10,000 Residents

The station-rate histogram shows that the cleaned station-per-10k rate is also right-skewed. Most tracts have zero or very low station availability, while a few have high per-capita station rates.

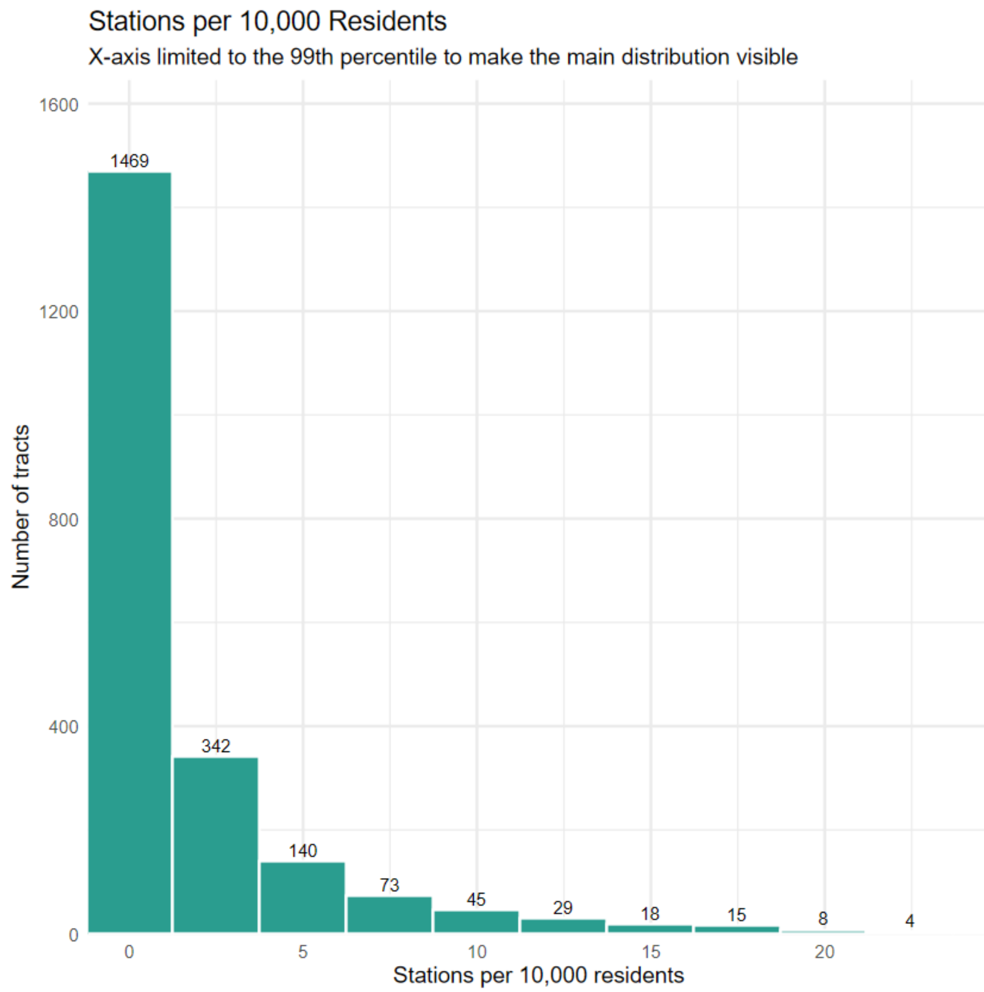

Figure 4: Distribution of cleaned stations per 10,000 residents.

7.5 Equity-Oriented EDA

Among tracts with population at least 500, tracts with at least one station have a median income of approximately $112,529, compared with $98,265 for tracts without a station. Tracts with stations also have lower median poverty rates, about 6.1% compared with 7.6%. No-vehicle household shares are similar, about 6.0% in tracts with stations and 6.4% in tracts without stations.

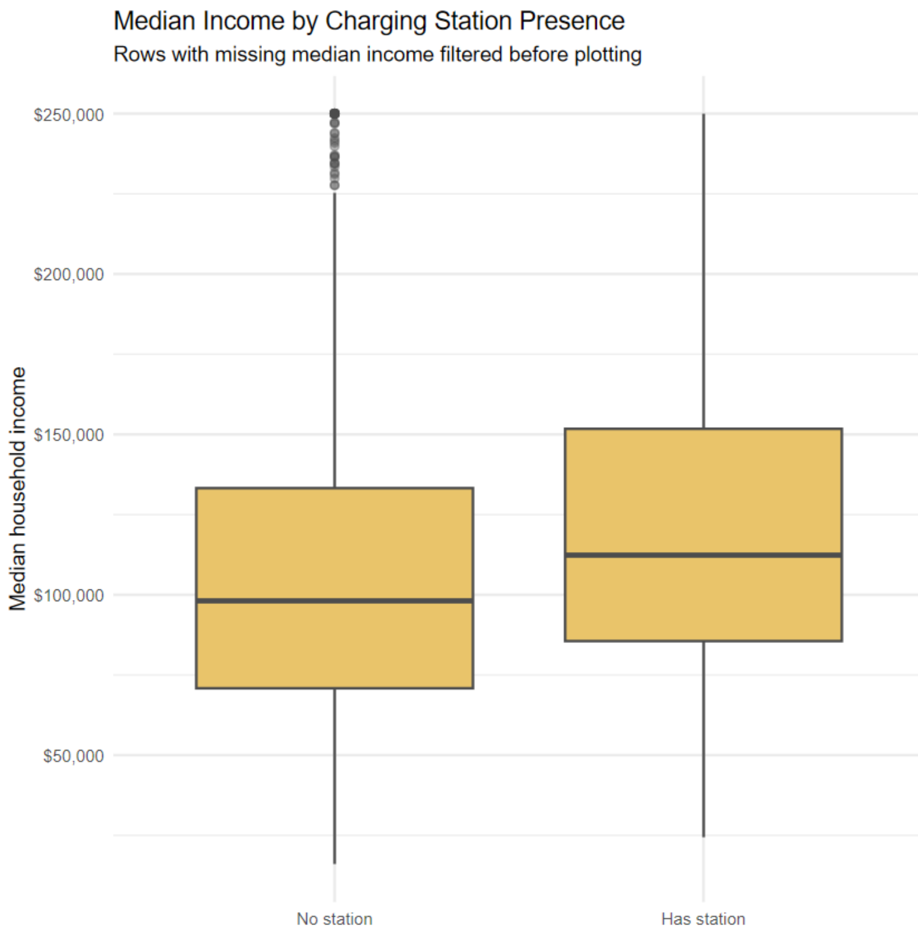

Figure 5: Median household income by station presence.

Based on the boxplots it suggest that station rate and demographic variables are not the most significant which suggests that no single demographic variable fully explains station access. Charger placement also depends on additional geographic and infrastructure factors.

8 Mapping and Spatial Interpretation

8.1 Map Color Choices

The maps use color scales selected to match the meaning of each variable:

- Charging availability maps use a sequential viridis scale. Brighter color means more stations per 10,000 residents.
- Median income uses a sequential scale. Brighter color means higher income.
- No-vehicle households use a sequential scale. Brighter color means a higher share of households without vehicle access.

- Distance to nearest charger uses a blue sequential scale. Darker blue means farther from the nearest charger, or lower spatial accessibility.
- Model residuals use a diverging blue-white-red scale centered at zero. Blue means fewer stations than predicted, white means close to predicted, and red means more stations than predicted.

This map-color logic is correct because variables such as income, station availability, and no-vehicle share are ordered quantities, while residuals have meaningful negative and positive directions.

8.2 Charging Availability Map

The raw map demonstrates why cleaning is necessary: a small number of low-population tracts create extreme per-capita rates. The cleaned map is the main version to interpret.

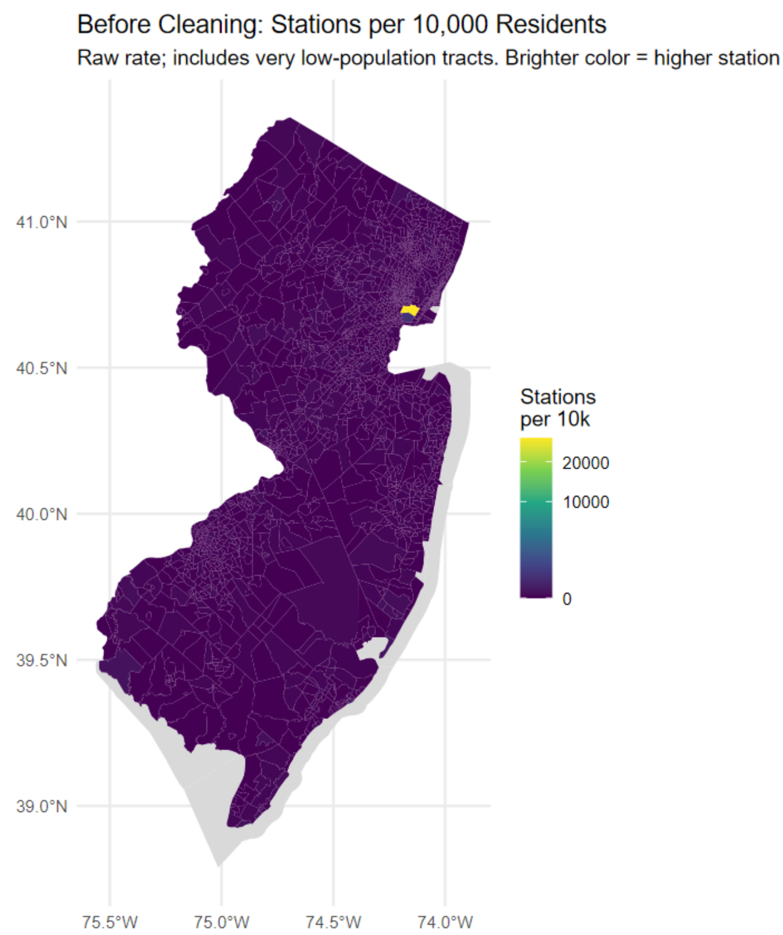

Figure 6: Raw stations per 10,000 residents before excluding low-population tracts. Extreme values can distort the color scale.

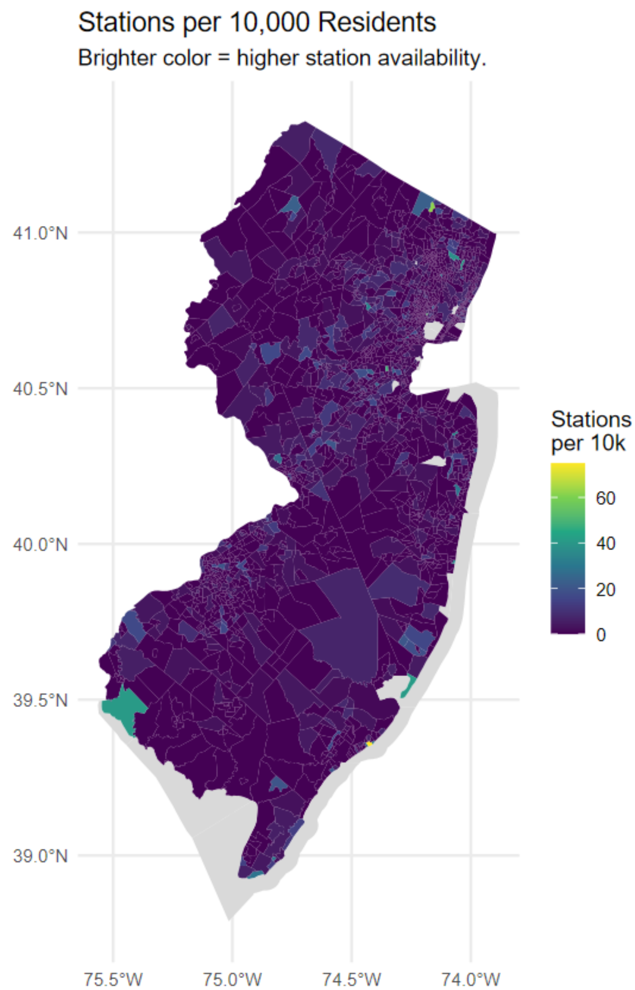

Figure 7: Cleaned stations per 10,000 residents. Tracts with population below 500 are excluded from the rate map, and values above 75 are capped for display.

The cleaned map suggests that public charging availability is not even and concentrated in selected tracts rather than uniformly distributed.

8.3 Context Maps

The income map provides socioeconomic context. It should not be treated as proof that income causes station placement, but it helps compare the spatial pattern of income with the spatial pattern of charging access.

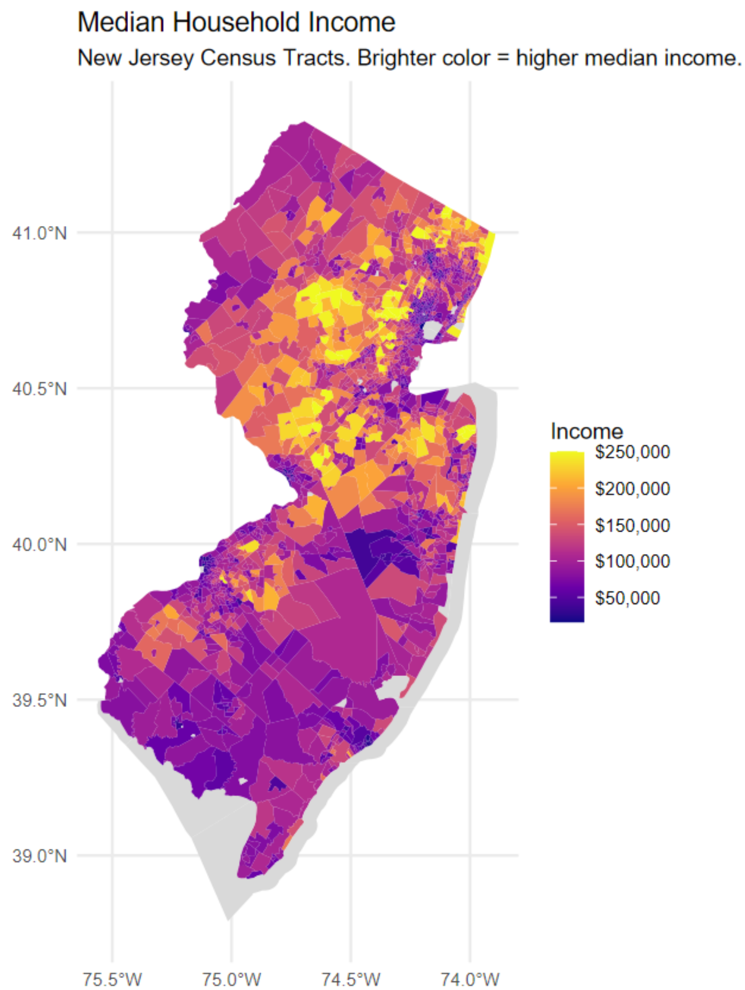

Figure 8: Median household income by census tract. Brighter color represents higher median household income.

The no-vehicle map provides an equity context. A higher no-vehicle share does not necessarily imply greater EV charging demand, but it helps identify places where transportation access and charging infrastructure intersect.

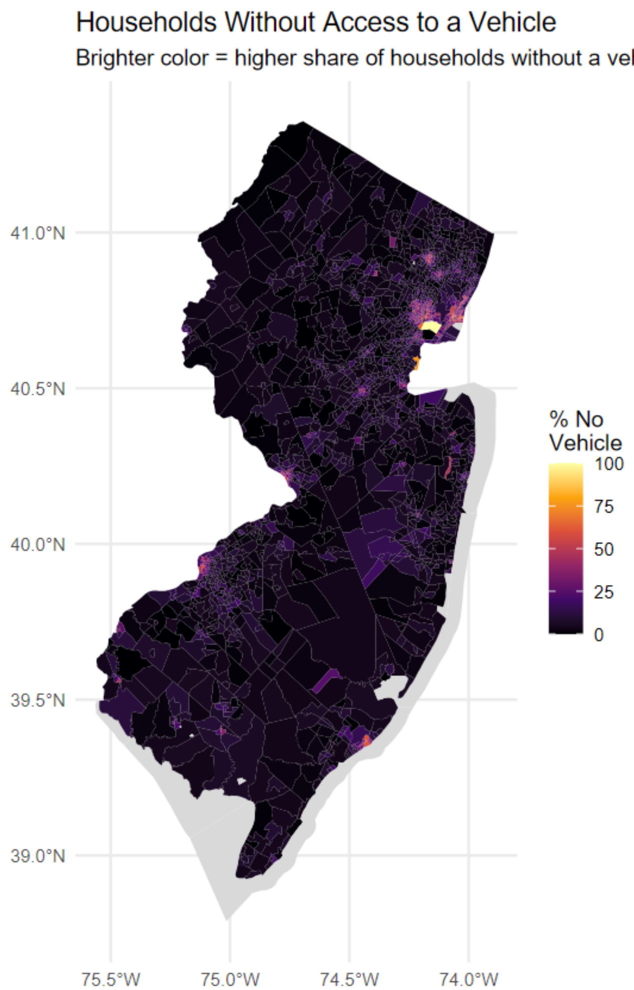

Figure 9: Percent of households without vehicle access. Brighter color represents a higher no-vehicle household share.

The station point map shows the raw spatial distribution of station locations.

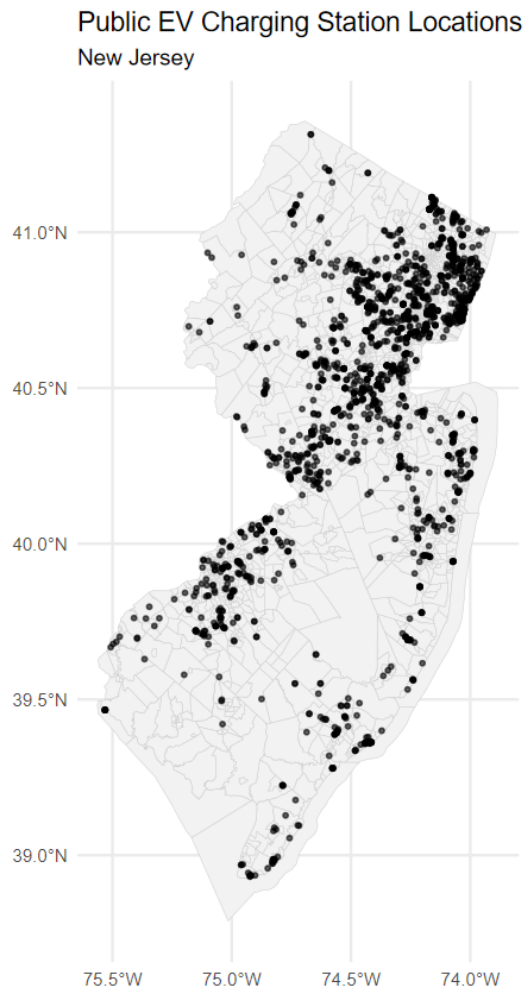

Figure 10: Public EV charging station locations over New Jersey tract boundaries.

8.4 Distance to Nearest Charger

Distance to the nearest charger is a spatial accessibility measure. It complements station counts because a tract can have zero stations but still be close to a charger in a neighboring tract.

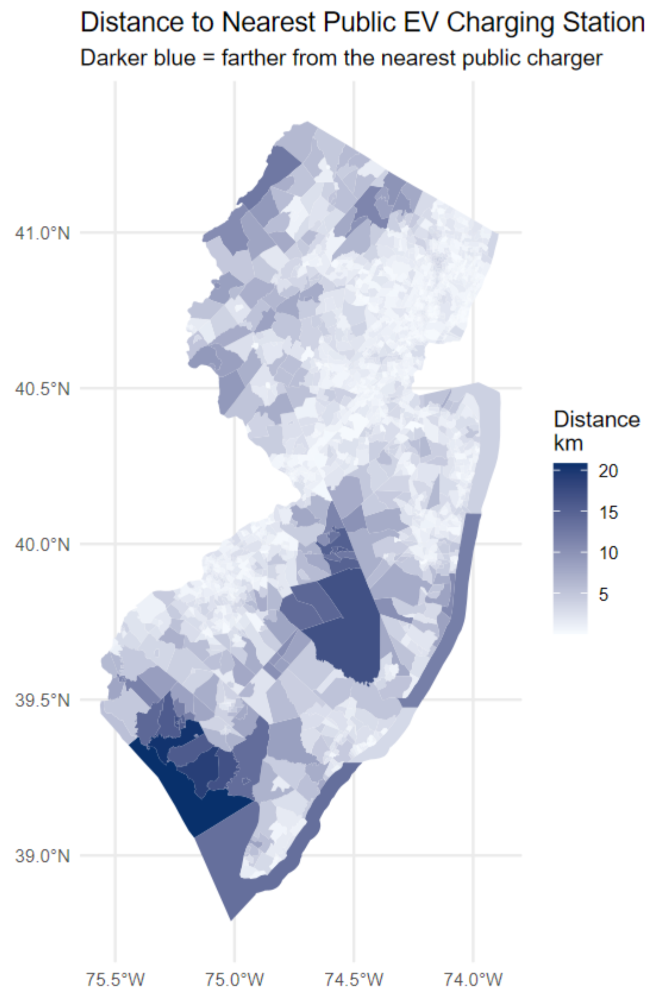

Figure 11: Distance from each tract representative point to the nearest public EV charger. Darker blue means farther away and therefore lower spatial accessibility.

The distance map is useful because it shifts the focus from "does this tract contain a station?" to "how far is this tract from the nearest station?"

9 Spatial Autocorrelation

Moran's I tests whether nearby tracts tend to have similar charging rates. The formula is:

$$I = \frac{n}{W} \frac{\sum_i \sum_j w_{ij}(x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2}, \tag{7}$$

where x_i is the charging access value in tract i, w_{ij} represents whether tracts i and j are neighbors, and W is the sum of all spatial weights.

The Moran's I result for cleaned station rates is:

Statistic	Value
Moran's I	0.0907
p-value	1.754×10^{-13}

Table 6: Global Moran's I for cleaned station rates.

This indicates statistically significant (modest) positive spatial autocorrelation. In other words, tracts with similar levels of charging access tend to be near one another. Charging access is not randomly scattered across the state.
The value is modest, so the result should not be overstated. A correct interpretation is:

> EV charging access in New Jersey shows modest but statistically significant spatial clustering.

10 Regression Modeling

10.1 Why Negative Binomial Regression?

The station count outcome is a nonnegative count variable. The EDA shows many zero-station tracts and a long right tail. This makes ordinary least squares inappropriate. A Poisson model is a possible starting point, but the data are overdispersed, so a negative binomial model is more appropriate.

The baseline model is:

$$Y \sim \text{Negative Binomial}(\mu, \theta), \tag{8}$$

$$\log(\mu_i) = \beta_0 + \beta_1\text{Income} + \beta_2\text{Poverty} + \beta_3\text{NoVehicle} + \beta_4\text{OwnerOccupied} + \log(\text{Population}). \tag{9}$$

The model includes log(Population) so that larger tracts are not treated the same as smaller ones. This allows station access to be interpreted relative to population size.

10.2 Baseline Model Results

The baseline model found:

Predictor	Coefficient	p-value
Median income per $10k	0.0376	0.00333
Poverty rate	-0.0081	0.3056
No-vehicle household share	0.0014	0.8009
Owner-occupied share	-0.0092	0.00132

Table 7: Baseline negative binomial model results.

The median-income coefficient is positive and statistically significant. Because the model uses a log link, exponentiating the coefficient gives a rate ratio:

$$\exp(0.0376) \approx 1.038. \tag{10}$$

This means that a \$10,000 increase in median household income is associated with about a 3.8% higher expected station count, holding the other variables and population constant.
The owner-occupancy coefficient is negative and statistically significant:

$$\exp(-0.0092) \approx 0.991. \tag{11}$$

This means that a one percentage point increase in owner-occupied housing is associated with about a 0.9% lower expected station count, holding the other variables and population constant.Poverty rate and no-vehicle household share are not statistically significant in the baseline model.

10.3 Residual Diagnostics

After fitting the baseline model, residuals are inspected to identify where the model predicts too many or too few stations.

- Positive residuals mean the tract has more stations than predicted.
- Negative residuals mean the tract has fewer stations than predicted.
- Residual maps show geographic patterns in model error.

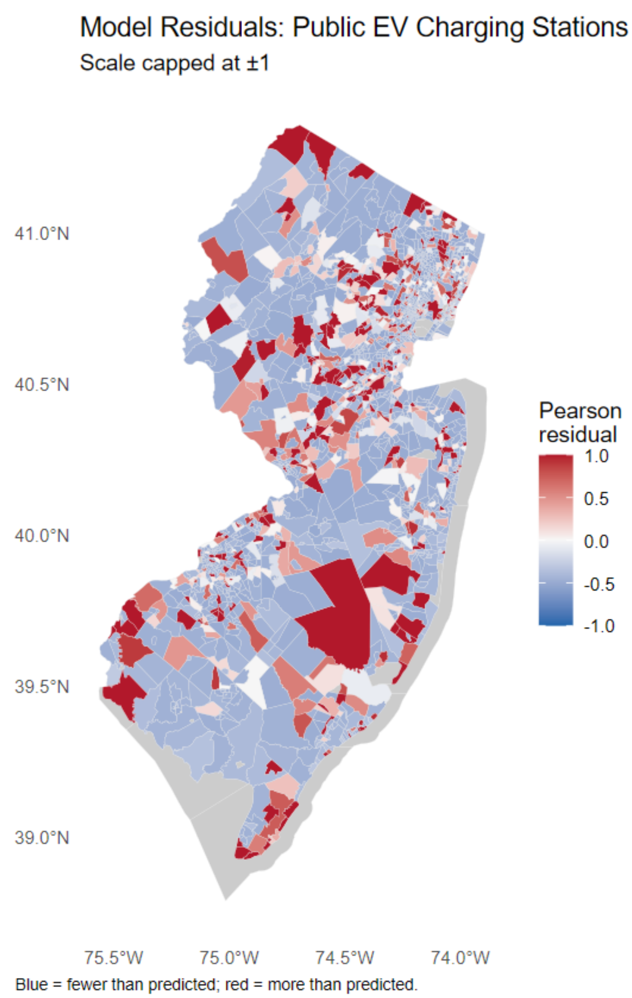

Figure 12: Baseline model residuals. Blue means fewer stations than predicted, white means close to predicted, and red means more stations than predicted.

The residual outlier table shows that some tracts have many more stations than predicted by the demographic model. These tracts likely represent commercial, highway, parking, workplace, or destination charging clusters.

Residual Moran's I is also significant:

Statistic	Value
Residual Moran's I	0.0952
p-value	1.523×10^{-14}

Table 8: Moran's I for baseline model residuals.

This means that the model residuals remain spatially clustered. The model is still missing geographic structure. This is an important limitation and supports the need for additional spatial predictors.

10.4 Density-Adjusted Model

Population density is added as a geographic control because station placement may depend on urban form, commercial activity, and travel demand.
The density-adjusted model is:

$$\begin{aligned}\log(\mu) = \beta_0 + \beta_1 \text{Income} + \beta_2 \text{Poverty} + \beta_3 \text{NoVehicle} \\ + \beta_4 \text{OwnerOccupied} + \beta_5 \log(1 + \text{PopulationDensity}) + \log(\text{Population}).\end{aligned} \tag{12}$$

The density-adjusted model found:

Predictor	Coefficient	p-value
Median income per $10k	0.0506	4.44×10^{-5}
Poverty rate	-0.0146	0.0632
No-vehicle household share	0.0116	0.0439
Owner-occupied share	-0.0213	2.67×10^{-12}
Log population density	-0.4193	$< 2 \times 10^{-16}$

Table 9: Density-adjusted negative binomial model results.

The density-adjusted model fits the data better than the baseline model. Its AIC decreases from about 5030 to 4950.9, and a lower AIC means better model fit.
The population density result should still be interpreted carefully. Since the model already adjusts for tract population, the density term is not simply asking whether more populated places have more chargers. Instead, it asks whether station availability per resident changes in denser tracts, after accounting for population and demographics.
The negative density coefficient may suggest that some charging stations are located in commercial, highway, or destination-oriented areas rather than in the most densely residential neighborhoods. It may also mean that the model is still missing important land-use variables.

11 Key Findings

1. Public EV charging access in New Jersey is uneven. Only about 32.1% of tracts have at least one public station.

2. Charging infrastructure is zero-heavy and right-skewed. Most tracts have no stations, while a small number contain many.

3. County-level totals show concentration in counties such as Bergen, Middlesex, Essex, Hudson, and Morris.

4. Moran's I shows modest but statistically significant spatial clustering.

5. Baseline regression indicates that higher income is associated with more stations, while owner-occupancy is associated with fewer stations.

6. The density-adjusted model fits better, but residual spatial autocorrelation remains.
7. Demographic variables explain some patterns, but geography and omitted land-use factors still matter.

12 Limitations

Important omitted variables include:

- Highway access;
- commercial land use;
- parking availability;
- shopping centers and employment centers;
- tourism and destination charging;
- utility and grid capacity;
- EV ownership and adoption rates;
- charging network and access restrictions.

The analysis uses census tracts, but charging access can cross tract boundaries. The distance-to-nearest-charger measure partially addresses this limitation, but network travel time would be better than straight-line distance.

13 Conclusion

Public EV charging infrastructure in New Jersey is uneven and spatially clustered. Although New Jersey has 1,822 public open EV charging stations and 6,027 total ports, only about one-third of census tracts contain at least one public station. EDA shows a zero-heavy and right-skewed distribution, meaning that most tracts have no stations while a small number have several. Moran's I confirms modest but statistically significant spatial clustering. Regression results suggest that higher-income tracts are associated with more public stations, while owner-occupied tracts are associated with fewer stations. Adding population density improves model fit, but residual spatial autocorrelation remains, showing that important spatial and land-use predictors are still missing.

Overall, the project provides evidence that public EV charging access is not evenly distributed across New Jersey neighborhoods. Demographics explain part of the pattern, but geography, transportation corridors, and land use likely play a major role.

References

[1] Alternative Fuels Data Center. *Alternative Fueling Station Data Download*. U.S. Department of Energy. Available at: `https://afdc.energy.gov/data_download`. Accessed May 5, 2026.

[2] U.S. Census Bureau. *American Community Survey 5-Year Data, 2024 ACS 5-Year Detailed Tables*. Available at: `https://api.census.gov/data/2024/acs/acs5`. Accessed May 5, 2026.

[3] U.S. Census Bureau. *TIGER/Line Shapefiles: 2024 New Jersey Census Tracts*. Available at: `https://www2.census.gov/geo/tiger/TIGER2024/TRACT/tl_2024_34_tract.zip`. Accessed May 5, 2026.